Title

ARGUS AI

• Intelligent Candidate Discovery & Ranking Engine*
"ARGUS AI is a production-inspired intelligent candidate discovery system based on modern retrieval,
reranking, signal engineering, and explainability principles."

ARGUS AI — SYSTEM ARCHITECTURE DECK

Redrob India Hackathon Submission — Confidential Slide 1 of 15

Problem Statement

• Massive Volumes: Recruiting teams are overwhelmed by processing hundreds of thousands of candidate profiles manually.
• Semantic Misalignment: Traditional keyword-based filters miss highly qualified candidates who describe their skills using
different vocabulary or conceptual synonyms.
• Data Quality Concerns: Candidate databases frequently contain suspicious, incomplete, or fraudulent data points (honeypots).
• The Hallucination Trap: Large Language Models (LLMs) used for resume scanning introduce hallucinated claims, high latencies, and high token costs.

ARGUS AI — SYSTEM ARCHITECTURE DECK

Redrob India Hackathon Submission — Confidential Slide 2 of 15

Proposed Solution

• Multi-Stage Selection Pipeline: Combines lexical precision and deep semantic matching to narrow 100k profiles down to the top 100.

• Deterministic Business Filters: Normalizes model outputs and injects strategic business signals (notice period, location target, company profile).
• Honeypot Gatekeeper: Scans profiles using rule-based algorithms to eliminate invalid entries.
• Deterministic Reason Builder: Generates factual, hallucination-free explanations within strict length constraints.

ARGUS AI — SYSTEM ARCHITECTURE DECK

Redrob India Hackathon Submission — Confidential Slide 3 of 15

Key Differentiators

• Zero-Hallucination Explainability: Natural text reasons are built deterministically from structured data, guaranteeing factual accuracy.

• Mathematical Scale Alignment: Pass Cross-Encoder logits through a Sigmoid mapping function before applying multiplier heuristics to ensure ranking consistency.

• Active Fraud Filtering: Incorporates a built-in Honeypot detection layer to filter out suspicious profiles before recommendations reach the recruiter.

• High Performance on CPU: Built and optimized for offline execution on standard CPU hardware without requiring expensive
GPUs.

ARGUS AI — SYSTEM ARCHITECTURE DECK

Redrob India Hackathon Submission — Confidential Slide 4 of 15

JD Understanding

• Docx Requirement Parser: Parses job descriptions to extract target job titles, core must-have skills, and secondary nice-to-have capabilities.
• Structured Query Construction: Compiles the parsed requirements into structured keyword lists for lexical matching and descriptive statements for semantic vector search.

ARGUS AI — SYSTEM ARCHITECTURE DECK

Redrob India Hackathon Submission — Confidential Slide 5 of 15

Ranking Methodology

• 1. Lexical Retrieval: Okapi BM25 scores keyword frequency to retrieve precise matches.
• 2. Semantic Retrieval: Dense vector indexing utilizing `all-MiniLM-L6-v2` embeddings in a FAISS index to find conceptual
matches.
• 3. Reciprocal Rank Fusion (RRF): Merges the lexical and semantic ranked lists using $k=60$ to combine relevance.
• 4. Cross-Encoder Reranking: Passes the top 300 candidates through `ms-marco-MiniLM-L-6-v2` to evaluate deep context-to-context semantic match.

ARGUS AI — SYSTEM ARCHITECTURE DECK

Redrob India Hackathon Submission — Confidential Slide 6 of 15

Explainability & Validation

• Factual Reason Templates: Automatically compiles candidate experience and skills into professional recruiter summaries (40-80 words).
• Rule Compliance: The output is validated using the official validator script, ensuring strict formatting, ordering, and candidate
count constraints are met.

ARGUS AI — SYSTEM ARCHITECTURE DECK

Redrob India Hackathon Submission — Confidential Slide 7 of 15

End-to-End Workflow

```text
Job Description (.docx)
↓
Requirement Parsing (Titles & Skills)
↓
Hybrid Search: BM25 Lexical + FAISS Semantic (RRF Fusion) ↓
Cross-Encoder Reranking (ms-marco-MiniLM-L-6-v2)
↓
Sigmoid Logit Scaling & Business Multiplier Scoring
↓
Honeypot Gatekeeper Fraud Filtering
↓
Template-based Explainability Reasoning
↓
Final Top-100 Verified Candidate Submission File

ARGUS AI — SYSTEM ARCHITECTURE DECK

Redrob India Hackathon Submission — Confidential Slide 8 of 15

```

ARGUS AI — SYSTEM ARCHITECTURE DECK

Redrob India Hackathon Submission — Confidential Slide 9 of 15

System Architecture

• FastAPI Backend: Hosts REST APIs for candidate ranking, profile analysis, and dashboard statistics.
• React + Vite Dashboard: An interactive recruiter interface with visual match grids, live queries, analytics charts, and submission downloader.
• Containerization: Docker and Docker Compose setups for reproducible builds and zero-setup local deployment.

ARGUS AI — SYSTEM ARCHITECTURE DECK

Redrob India Hackathon Submission — Confidential Slide 10 of 15

Results & Performance

Key Metrics

• Total Candidate Pool: 100,000 candidates
• Embedding Dimension: 384
• Top-K Retrieval: 2,000
• Cross-Encoder Top-K: 300
• Final Candidates: 100
• Pipeline Runtime: <15 sec
• FAISS Build Time: 0.19 sec
• Unit Tests: 14/14 Passed
• Submission Validator: Passed

ARGUS AI — SYSTEM ARCHITECTURE DECK

Redrob India Hackathon Submission — Confidential Slide 11 of 15

Technologies Used

• ML & Search: FAISS, Sentence-Transformers, Rank-BM25, PyTorch
• Backend Web APIs: FastAPI, Uvicorn, Pydantic, Loguru
• Frontend UI: React 19, TypeScript, Vite 8, Tailwind CSS v4, Recharts, Lucide Icons
• Infrastructure: Docker, Docker Compose, GitHub Actions CI

ARGUS AI — SYSTEM ARCHITECTURE DECK

Redrob India Hackathon Submission — Confidential Slide 12 of 15

Dashboard Demonstration

• Recruiter KPI Panel: Visualizes match percentages, location maps, and notice periods.
• Live Query Sandbox: Paste any job description to run the complete hybrid ML ranking pipeline instantly.
• Explorer Grid: Expand candidate details, review the breakdown of business multipliers, and read generated reasoning text.

ARGUS AI — SYSTEM ARCHITECTURE DECK

Redrob India Hackathon Submission — Confidential Slide 13 of 15

Future Scope

• Database Migration: Scale indexing to `pgvector` in PostgreSQL for enterprise-level persistence.
• Incremental Indexing: Support real-time indexing of newly added resumes.
• Resume Uploader: Add PDF parsing to ingest candidate profiles directly.
• Recruiter Feedback Loop: Incorporate online learning-to-rank using human-in-the-loop decisions and label feedback.

ARGUS AI — SYSTEM ARCHITECTURE DECK

Redrob India Hackathon Submission — Confidential Slide 14 of 15

Conclusion

• Proven Precision: ARGUS AI effectively separates the top talent from 100k records in less than a minute.
• Compliant & Reliable: 100% test pass rate and official submission validation passed.
• Recruiter Centric: A seamless interface combining sophisticated search science with actionable business signals.

ARGUS AI — SYSTEM ARCHITECTURE DECK

Redrob India Hackathon Submission — Confidential Slide 15 of 15